Kapasa Makasa University

Information Communication Technology Department
(Cyber Security)

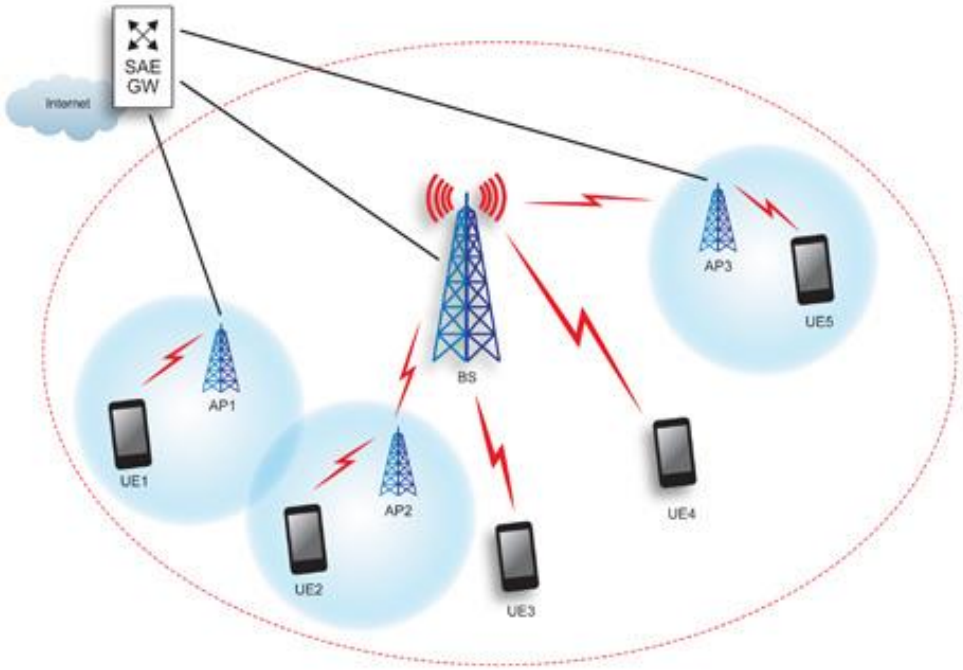
Wireless Security

(CYS331)

Lecture 4

BY Z KALUBA

Wireless Transmission Media



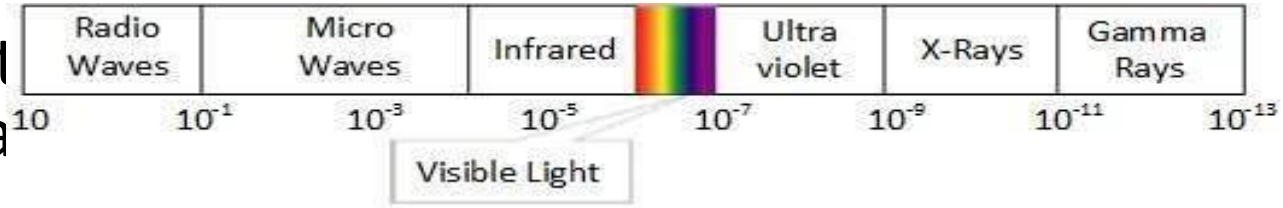
Wireless Transmission

- Wireless transmission is a form of unguided media. Wireless communication involves no physical link established between two or more devices, communicating wirelessly. Wireless signals are spread over in the air and are received and interpreted by appropriate antennas.
- Wireless communication can be used for cellular telephony, wireless access to the internet, wireless home networking, and so on.

Wireless Transmission

- When an antenna is attached to electrical circuit of a computer or wireless device, it converts the digital data into wireless signals and spread all over within its frequency range. The receptor on the other end receives these signals and converts them back to digital data.

➢A little part for wireless tra



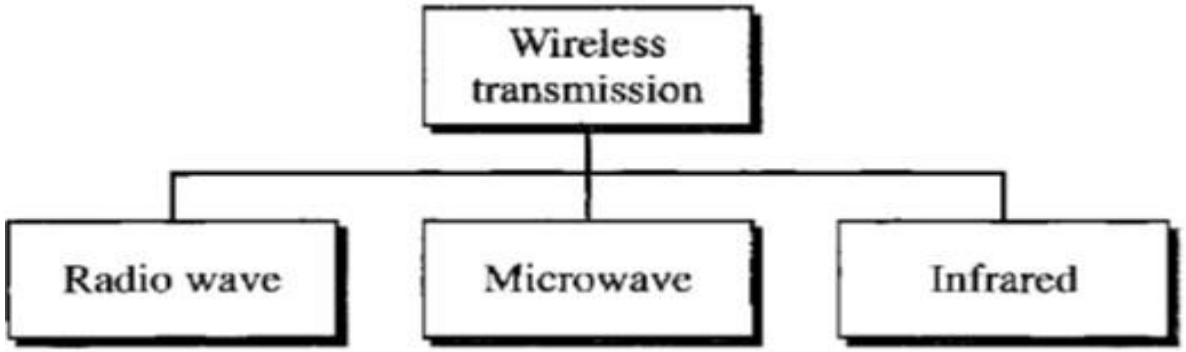
Generation of Wireless Communication

The different generations of wireless communication include the following.

- 1st Generation (1G)
- 2nd Generation (2G)
- 3rd Generation (3G)
- 4th Generation (4G)
- 5th Generation (5G)

Wireless Transmission

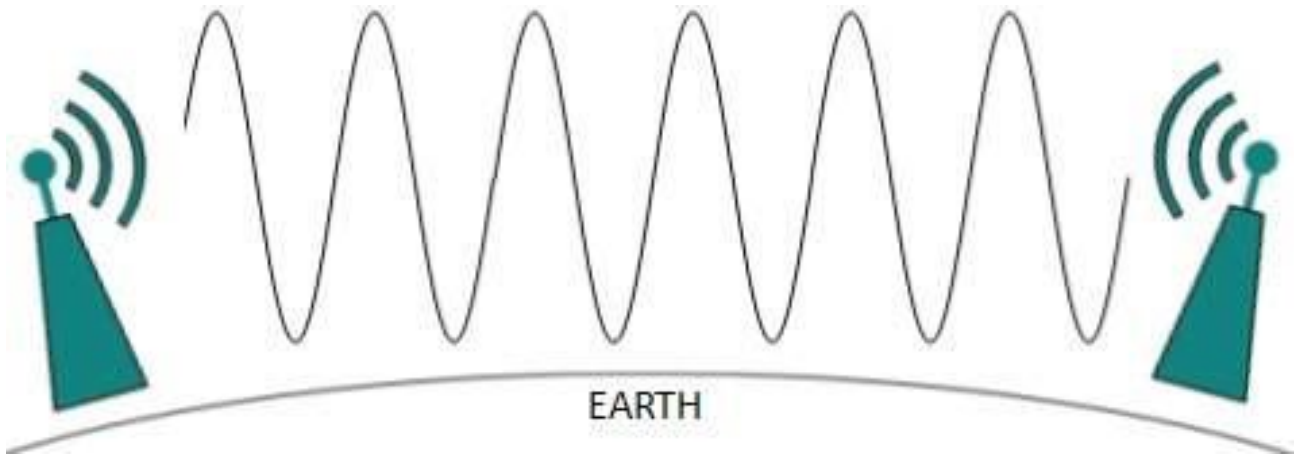
- Types of Wireless transmission.



Applications of Wireless Transformation

- Applications of wireless communication involve security systems, television remote control, Wi-Fi, Cell phones, wireless power transfer, computer interface devices, and various wireless communication-based projects.

Radio Wave



Radio Wave

- A radio wave is an electromagnetic phenomenon which travels through space and through many nonconductive/nonmagnetic materials. Radio waves are a part of the broad electromagnetic spectrum.
- They range from 3 KHz to 3000 GHz. These radio waves are further classified into the following:

➢ Extremely Low Frequency - 3 to 30 Hz. Super Low Frequency - 30 to 300 Hz. Ultra Low Frequency - 300 to 3000 Hz. Very Low Frequency - 3 to 30 KHz. Low Frequency - 30 to 300 KHz.

➢ Medium Frequency - 300 to 3000 KHz.

➢ High Frequency - 3 to 30 MHz. Very High Frequency - 30 to 300 MHz. Ultra High Frequency - 300 to 3000 MHz. Super High Frequency - 3 to 30 GHz. Extremely High Frequency - 30 to 300 GHz. Tremendously High Frequency - 300 to 3000 GHz.

Advantages of Radio wave

- Speed: electromagnetic waves of any kind are the fastest things in the universe.
- Signal-to-noise ratio: radio waves can be used to communicate with deep space probes. Visible or infrared radiation is not up to the task because such a communication system would have to outshine the sun.
- High power is readily available: The magnetron in a microwave oven is inexpensive and can produce about 1 kW of output. Longwave broadcast and communications transmitters have been built that produce over a million Watts. There are frequencies in the electromagnetic spectrum, such as the “tera Hertz gap,” where it is not (yet) feasible to produce this kind of power.
- Can penetrate obstacles that are opaque to visible light and infrared.
- Low photon energy is safer for living beings than ultraviolet radiation or x-rays.

Disadvantages of Radio wave

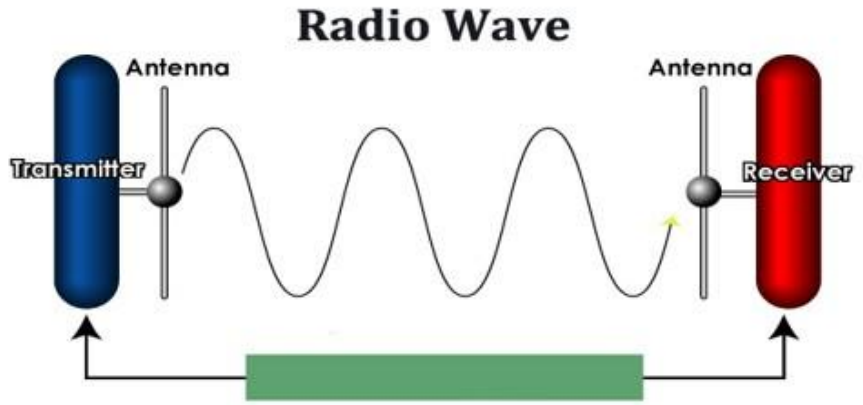
- Relatively low bandwidth: Infrared and visible lasers can now be modulated at much faster rates, making fiber optics the medium of choice for high throughput digital links.
- Long wavelengths: Directional antennas and reflector dishes become quite large and heavy compared to lenses designed for visible light.
- Long wavelengths: It is much harder to create a highly directional beam (in the way that a LASER is highly directional).
- Long wavelengths: Resolution is very limited when this type of radiation is used for imaging purposes. 430 MHz can take really great pictures of craters on the moon, but it can’t take good pictures of cavities in your teeth.

Characteristics of a Radio wave

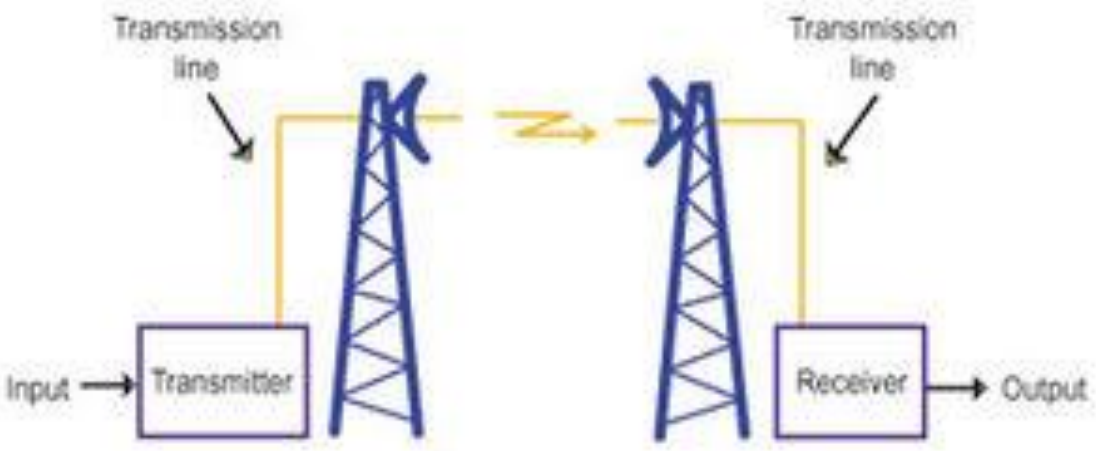
- Radio wave is a method of transmitting intelligence from one location to another by means of electromagnetic radiation.
- Radio waves can travel through long distances and penetrate through materials unlike light and heat waves.
- Radio waves are invisible.
- Radio waves are normally not felt.
- Radio waves are highly regulated and omni directional antennas.
- Radio waves have frequencies as high as 300 gigahertz to as low as 30 hertz.

Application of radio wave

- Radio waves carry signal for AM and FM radio.
- It is highly use in cellular telephones and Television.
- Radio waves are useful for Police and Radar.
- Radio waves is used in magnetic resonance imaging or MRI to produce pictures of tissue.



Microwave


Transmission line
Transmission line
Input
Transmitter
Receiver
Output

Microwave

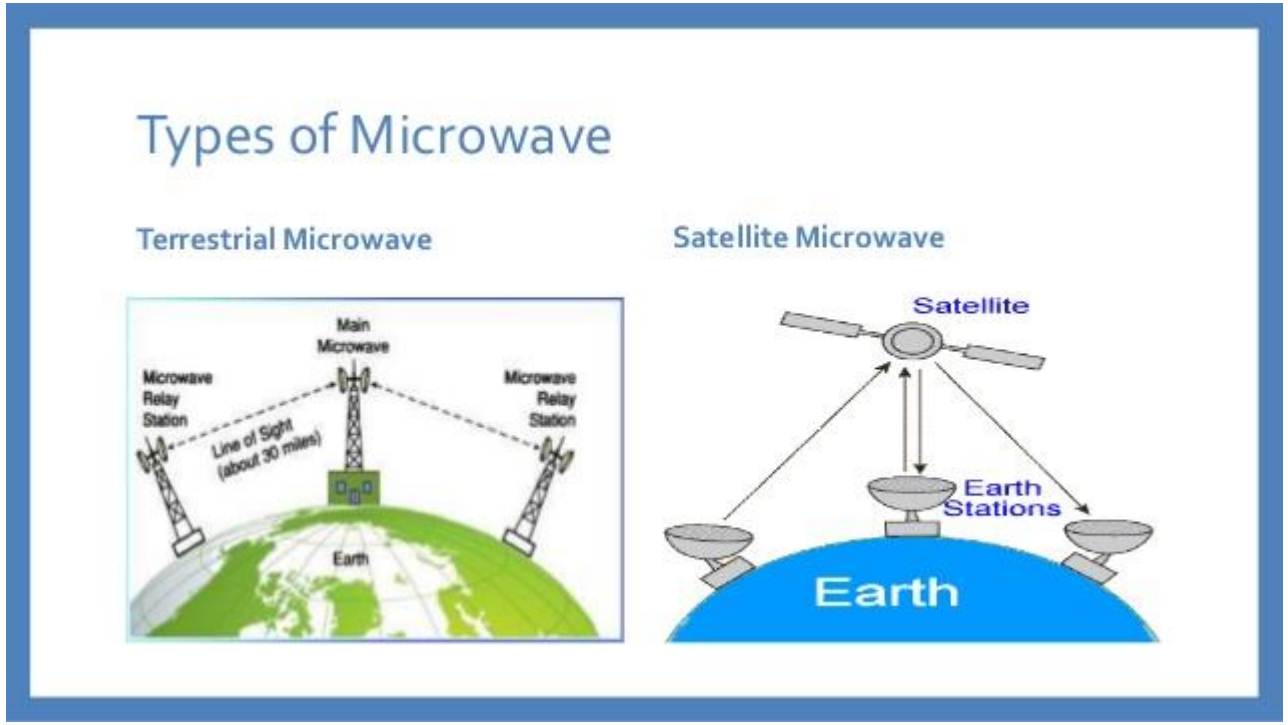
- Electromagnetic waves above 100 MHz tend to travel in a straight line and signals over them can be sent by beaming those waves towards one particular station. Because Microwaves travels in straight lines, both sender and receiver must be aligned to be strictly in line-of-sight.

- Microwaves can have wavelength ranging from 1 mm – 1 meter and frequency ranging from 300 MHz to 300 GHz.

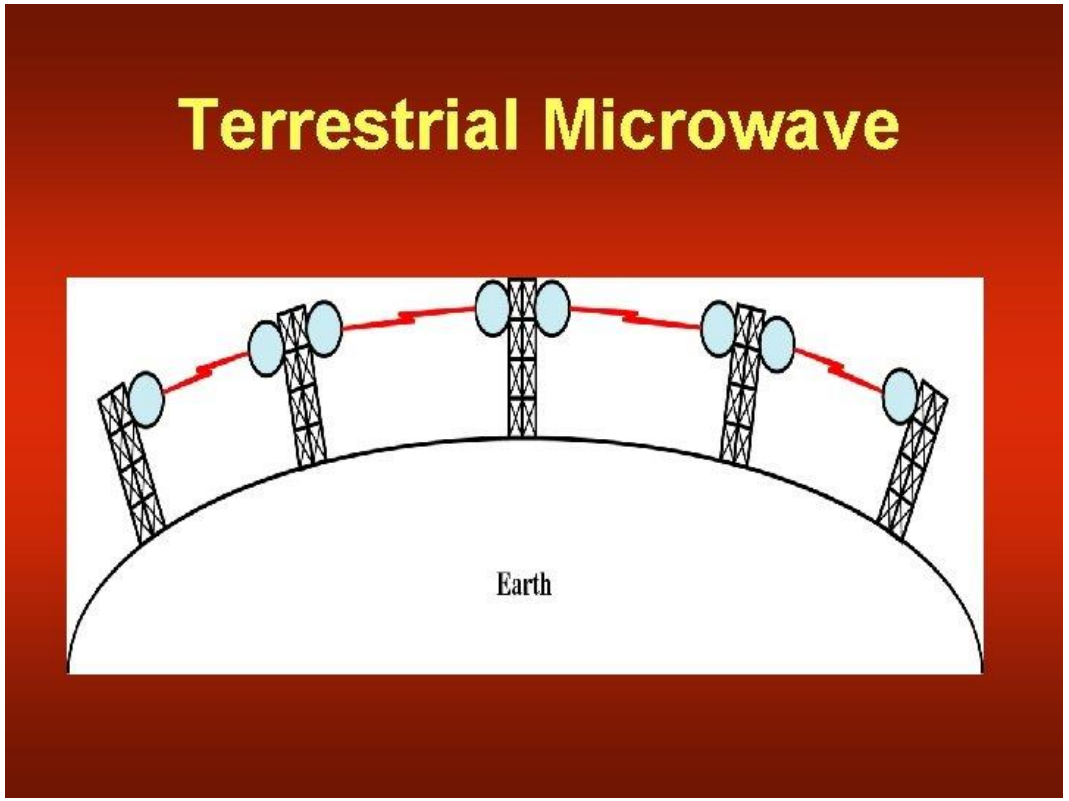
- Microwave transmission depends highly upon the weather conditions and the frequency it is using.

- Microwaves are used for unicast communication such as cellular telephones ,satellite networks and wireless LANS.

There are two types of Microwave:



Terrestrial microwave

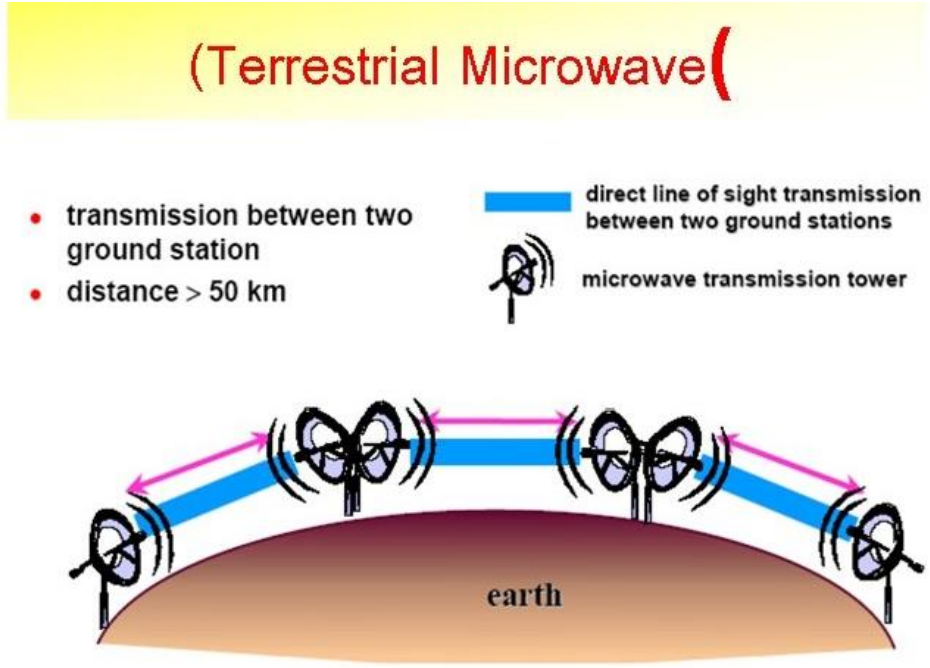


Terrestrial microwave

- Terrestrial microwave is a wireless microwave networking technology that uses line-of sight communications between pairs of Earth-based transmitters and receivers to relay information.

- Terrestrial microwave transmissions are sent between two microwave stations on the earth (earth station). It is the most common form of long-distance communication.
  Satellite microwave transmissions involve sending microwave transmissions between two or more earth-based microwave stations and a satellite.

Terrestrial microwave



Characteristics

- Fewer Repeaters but requires line of sight transmission.
- Higher frequencies give higher data rates (Common frequencies used for transmission are in the range 1 to 40 GHz).
- Main source of loss is attenuation caused mostly by distance ,rainfall and interference.

Applications

- Used for long-distance telephone service.
- Parabolic dish transmitter, mounted high.
- Used for both Voice and TV transmission
- Used by common Carriers as well as private networks.

Satellite Microwave



Satellite Microwave

- A Communication satellite is a microwave relay station .Used to link two or more ground stations.
- Satellite microwave Transmission system uses satellites for broadcasting and receiving of signals . These systems need satellites which are in the geostationary orbit which is 3600 km above the earth

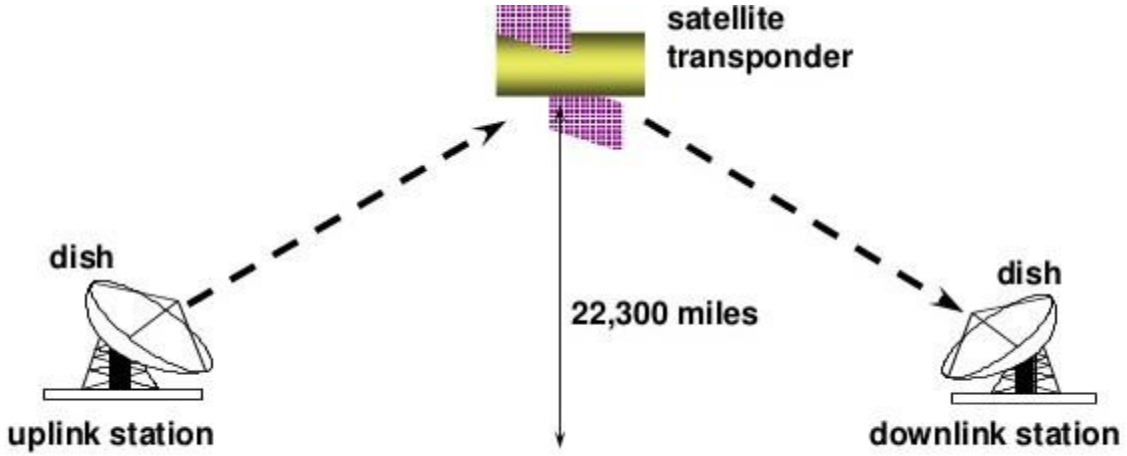
- Advantage:
  - No cables need
  - Multiple channels available
  - Wide bandwidth

- Disadvantage:
  - Expensive towers and repeaters
  - Subject to interference such as passing airplanes and rain
  - Line of sight requirement

Characteristics

- The optimum frequency range for satellite transmission is 1 to 10 GHz
    I. Lower has significant noise from natural sources
    II. Higher is attenuated by atmospheric absorption and precipitation.
- Satellites use a frequency bandwidth range of 5.925 to 6.425 GHz from earth to satellite(uplink) and a range of 3.7 to 4.2 GHz from satellite to earth(downlink)

Satellite Transmission Process



Applications of Satellite Microwave

- Private Business networks
  I. Satellite providers can divide capacity into channels to lease to individual business users.
- Television distribution
  I. Programs are transmitted to satellite then broadcast down to number of stations which then distributes the programs to individual viewers
  II. Direct Broadcast Satellite transmits video signals directly to the home user

Applications of Satellite Microwave

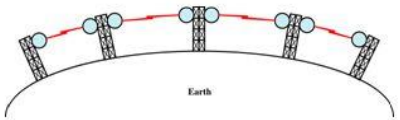
- Long distance telephone transmission
    I. High-usage international trunks
- Global positioning
    I. Global Positioning System (GPS)

Difference between terrestrial and satellite microwave

Microwave

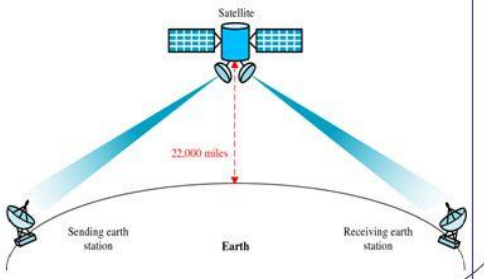
Terrestrial Microwave

- Parabolic dish
- Focused beam
- Line of sight
- Long haul telecommunications
- Higher frequencies give higher data rates



Satellite Microwave

- Satellite is relay station
- Satellite receives on one frequency, amplifies or repeats signal and transmits on another frequency
- Requires geo-stationary orbit
  - Height of 35,784km
- GEO vs. LEO
- Frequency Bands: C, Ku, and Ka
- Television
- Long distance telephone
- Private business networks



Infrared



Infrared

- Infrareds (IR) transmission involves sending light signals at a frequency between visible light and radio waves. Commonly used in TV remote controls, now is used to provide LAN connections.
- It is a line-of-sight transmission and has a maximum coverage of 30 to 80 feet.
- Increasingly, computers and devices such as printers come with IrDA ports, which enable the transfer of data without the use of cables.

Characteristics

- *Infrared communications utilize transmitters and receivers that sending signals through infrared light .*
- *Infrared operates by sending a signal in a straight line from the source to the destination where frequencies just above microwaves are used .*
- *If this "direct line of sight" signal is obstructed by an object, then the signal will not transmit as the signal cannot penetrate through the surface of the object, however it can reflect from surfaces such as the ceiling of a room .*
- *It is generally used for short distance transmission such as a TV remote .*
- *It does not require an antenna to operate.*

Applications

- Remote controls for television, stereos and other home appliances.
- Wireless LANs
- Wireless modem, keyboard, mouse, printer etc
- Fire detectors
- Night vision systems
- Intrusion detection systems
- Motion detectors

END